DB-Curator

Agentic database construction for scientific evidence

A tool that searches literature, proposes candidates, and verifies each database row with auditable evidence

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The task: build a database, not just answer a question

Scientific pain point

Specialized scientific databases are still mostly built by expert literature curation.

The hard part is not finding chemical names; it is deciding whether each name belongs in the database.

Release requires identity, topic scope and a paper trail that another scientist can inspect.

Agentic framing

The literature set is built during the run: searches change as candidates and evidence gaps appear.

Each candidate becomes an evidence contract: required predicates must pass and exclusion contexts must fail.

The system stops with an explicit state: supported, rejected, or evidence-limited.

What users get

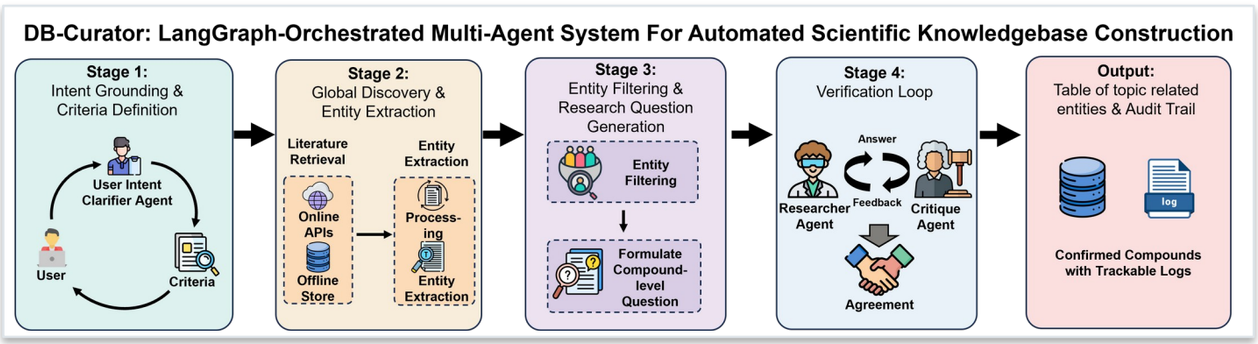
Curated compound rows with identifiers, topic labels and decision states.

Cited papers plus quote or local-context support for the decision.

A compact audit trail so humans review hard cases instead of searching from scratch.

One-sentence story: DB-Curator replaces open-ended manual literature curation with an agentic workflow that searches, verifies and leaves an audit trail.

How the workflow connects search to verification



1. Dynamic literature

Broad search first; targeted search follows when a candidate or missing predicate needs evidence.

2. Candidate grounding

Ground chemical names, then ask: does this compound fit the database definition?

3. Predicate verification

Researcher finds evidence; Critique checks whether required predicates are supported.

How to read this figure: left side builds the candidate pool; right side checks whether each candidate satisfies the rubric.

supported | unresolved | exclusion | exhausted

What the system produces: release-ready rows

246/250

release-accepted rows

Overlap plus human-accepted non-overlap rows.

178

benchmark overlaps

Known database entries recovered automatically.

68

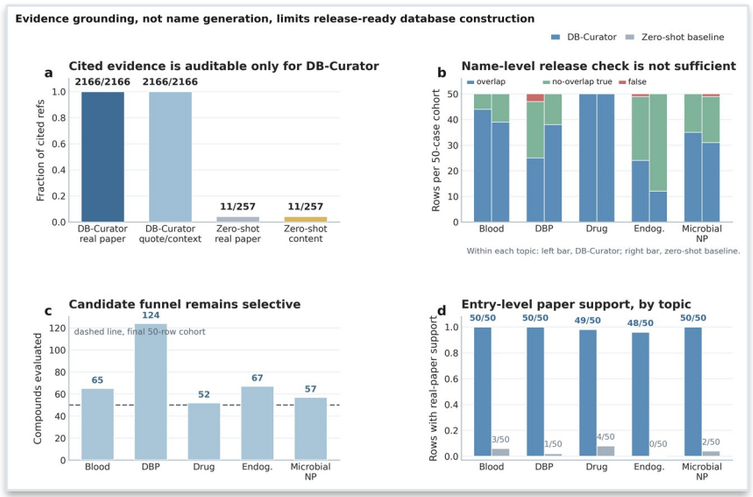
new valid entries

Absent from benchmark, accepted by human experts.

4

errors

False or out-of-scope rows.



How to read the figure: DB-Curator is strong because accepted names also carry auditable paper evidence.

Why this matters scientifically

Benchmarks are useful but incomplete for open-ended database building.

A useful system should recover known compounds and also surface evidence-backed additions.

The human role shifts from searching everything to auditing the few non-overlap rows and their evidence.

How every accepted output is verified

Paper-level audit

First ask: is the cited item a real primary paper? Resolve it using DOI, PMID, title or metadata.

Then ask: does the paper actually contain the quoted span or equivalent local context?

Compound lookup pages are useful for identity, but they are not counted as primary-paper evidence.

Predicate-level audit

A paper must support the actual rubric predicate, not just mention a related chemical or context.

Reagent, solvent and out-of-scope contexts block release even if the compound name is real.

If evidence is missing, the agent searches again or leaves the row evidence-limited.

2,166/2,166

fixed-cohort support

Refs with quote/context support.

7,430/7,453

benchmark-scale support

Auditable refs across five domains.

11/257

zero-shot baseline

Usable refs from direct generation.

Presenter cue: stress that the output is not trusted because the model says so; it is trusted because the cited paper and rubric predicate can be inspected.

What this shows about agentic AI for science

1,887

evaluated candidates

Scale of evaluation across five domains.

977

system Confirmed

Rows judged supported by the verification loop.

49

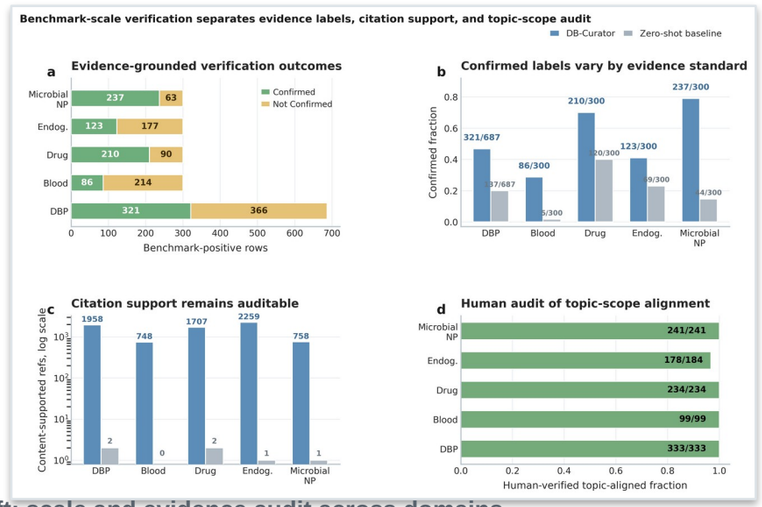
rescued by critique

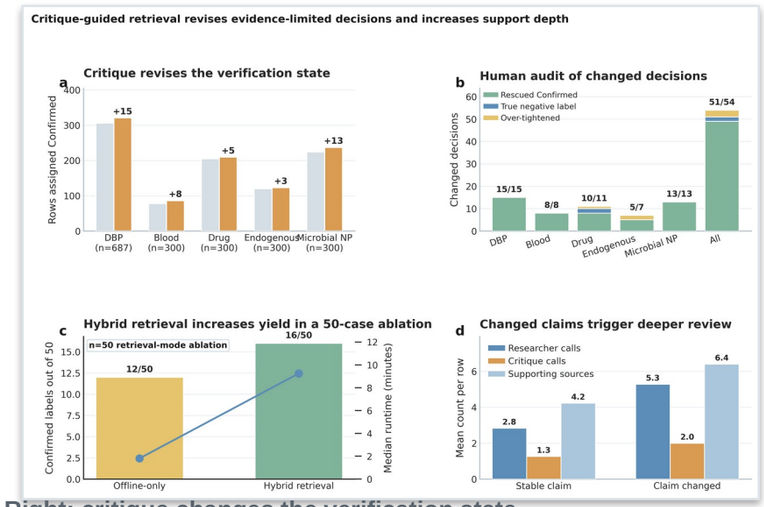
Critique found the missing evidence path.

5

tightened

Critique prevented over-acceptance.





Left: scale and evidence audit across domains.

Right: critique changes the verification state.

Closing message: the agentic contribution is not just more retrieval; it is dynamic evidence collection plus predicate-level stopping and correction.